

FieldTrip: An Open Source MATLAB Toolbox for Biomedical Signal Analysis

JAI SACHITH PAUL

In previous EFY Plus issue we had discussed BioSig Library for biomedical signal processing. Now, let us take a look at yet another toolbox for similar applications.

FieldTrip is an open source MATLAB toolbox that helps in the analysis of non-invasive and invasive electrophysiological data including magnetoencephalography (MEG) and electroencephalogram (EEG) signals.

The software provides a common platform for both method developers as well as the scientific community who work on biomedical signals. Various signal-processing techniques including data pre-processing, event-related field/response analysis, parametric and non-parametric spectral analysis, forward and inverse source modelling, connectivity analysis, classification, real-time data processing, and statistical inference and distributed computing is possible with the help of FieldTrip toolbox. The software, consisting of more than 100,000 lines of code, defines more than 100 high-level functions and nearly 1000 low-level functions that are distributed freely under GNU public licence.

What FieldTrip can offer

The research article titled 'FieldTrip: Open Source Software for Advanced Analysis of MEG, EEG and Invasive Electrophysiological Data' by Robert Oostenveld and others discusses the structure and main functionalities of the toolbox.

fileio. Main functions of this module include acquisition of electrophysiological data from various sensors, sorting information in received

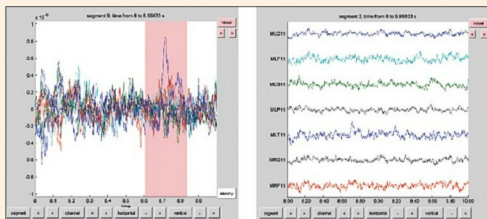


Fig. 1: Data visualisation using functions from FieldTrip toolbox

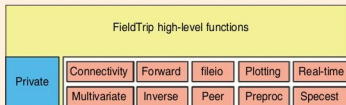


Fig. 2: The toolbox structure

data into header information, events (such as triggers), actual recorded signals and the like.

Preproc. Various pre-processing tasks associated with raw data are accomplished with the help of this module. This includes time-domain filtering operations, re-referencing, base line correction and much more.

Specest. This module allows the user to implement various spectral analysis methods. Techniques like frequency decomposition and wavelets could be used for estimating the power and phase of oscillatory components.

Connectivity. FieldTrip connectivity module focuses on connectivity in the frequency domain. The widely-used connectivity matrices such as coherence, phase slope index and partial directed coherence are available in this module.

Forward. Various algorithms that

provide solutions to the forward problem in MEG and EEG are implemented in this module.

Inverse. In order to estimate the location of neuronal ac-

tivity and its strength, various source-reconstruction algorithms have been implemented in this module. These include dipole fitting based on non-linear optimisation, linear estimation of distributed source models and others.

Multivariate. The researcher could make use of various machine-learning classification algorithms in this module for offline single-trial analysis as well as online brain computer interface (BCI) applications.

Plotting. Functions that help in the visualisation of complex data including source reconstruction and multichannel decomposition of time and frequency are available in this module.

Real-time. The module facilitates reconstruction of real-time BCI systems. This is accomplished with the help of a FieldTrip buffer that allows the acquisition client to stream data in